
APMTH 50 Problem Set 4

Due on Friday, April 7, by 5pm

Collaboration policy

Collaboration on homework and labs is encouraged, but each student must write solutions in their own words, and not copy them from anyone else. Please write the name(s) of your collaborator(s) at the top of the homework.

Submission guidelines

Please upload your homework as a single Word or pdf document. **For this homework, you should submit all of your code as well as the output from your code.**

Announcements

Please make sure to schedule a time to meet with me to discuss your project. You must do this by 5pm on April 6.

Problems

1. Write a half page description of your project proposal, based on the conversation you had with me. Your proposal should include:
 - (a) A specific question that your project will attempt to answer.
 - (b) An outline of steps you plan to take to answer the question. It would be good you to list a few benchmarks that you plan to meet on your way to answering the main question.
 - (c) To be clear, the project you turn in on May 3 may not fully answer the question (a) that you set out to address. This is fine and normal and you try not to be discouraged. We are much more interested your *modeling process*, ie, how you defined the problem and formulated it mathematically/computationally, and your thoughtful discussion and critique of your model assumptions and results. You will primarily be graded on your attempt at this process, irrespective of the results. See the grading rubric in the class 12 slides.
2. **Channel entropy.** Consider a channel where information is sent as three-digit decimal numbers of the form 000, 001, 002, $\dots$, 148, 149. Assume that each of the 150 numbers is equally probable. The entropy of a number is therefore $H_N = -150(\frac{1}{150} \log_2 \frac{1}{150}) = \log_2 150$.
 - (a) Calculate the Shannon entropy of the first digit, second digit, and third digit.
 - (b) Calculate the sum of the entropies of the three digits. In words, explain why you think the sum is not equal to H_N .

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- (c) Suppose that the second digit of a number is 3. What are entropies of the first and third digits? Are the entropies the same as in part (a)?

3. **Digram analysis of different languages.** In his paper, Shannon considers the digram structure of English, where he looks at the probabilities of two-letter combinations. It turns out that digrams represent a simple but surprisingly powerful way to differentiate between languages. To illustrate this, we have downloaded out-of-copyright books in five different European languages from [Project Gutenberg](#):

0. English: *The Voyage Out* by Virginia Woolf (1915)
1. French: *L'Atlantide* by Pierre Benoît (1920)
2. German: *Siddartha* by Hermann Hesse (1922)
3. Italian: *Dal Cellulare a Finalborgo* by Paolo Valera (1899)
4. Spanish: *La Voz de la Conseja* edited by Emilio Carrère (1920)

These books are posted in the information theory homework folder on the course website. They have been pre-processed to remove any special characters and accented characters. A program `digram.m` is supplied, which scans each book and calculates the probability distribution of the digrams. It removes punctuation, converts everything to lower case, and then searches for occurrences of every possible digram. In the analysis, it considers 27 characters, with $1 = \text{a}$, $\dots$, $25 = \text{y}$, $26 = \text{z}$, and 127 corresponding to a space. After the program has run, it will have assembled probabilities $p_{i,j}^k$ of the occurrence of digram (i, j) in language k .

- (a) Run the program `digram.m`. It will build the digram probabilities and then output a 2D matrix for the English digrams. What do you notice about the row for the letter `q`? Comment on a few other features of the matrix.
- (b) Modify the program to plot the difference between the Italian and Spanish digram matrices. You may (or may not) want to write a function to do this. Comment on some key differences between the two languages.
- (c) Add lines to the program to calculate the entropy of selecting a random digram for each of the five languages. What language has the most entropy? What language has the least? Compare the entropies against a hypothetical language in which all 26×26 digrams are equally probable. *You will need to be careful when calculating the entropy since some of the entries in the digram matrix are 0.*
- (d) Add lines to the program to calculate a measure of the difference between two languages,

$$D(k, l) = \sqrt{\sum_{i=1}^{26} \sum_{j=1}^{26} (p_{i,j}^k - p_{i,j}^l)^2}.$$

Again, you may want to write a function to do this. What pair of languages are the most similar? What pair of languages are the most different?